# MeshCore Setup Guide

MakerHawk 2-pack ESP32 LoRa V3 (Heltec V3-compatible, SX1262, 915 MHz) · Two boards as phone radios that message each other

## What you need

- Both kits (board, 915 MHz antenna, 1100 mAh battery, case)
- A USB-C **data** cable (charge-only cables won't work)
- A Windows, Mac or Linux computer with **Chrome or Edge**
- The **MeshCore** app on your phone (Google Play or App Store)

## 1. Assembly (each board)

1. Snap the 915 MHz LoRa antenna onto the small round **U.FL/IPEX** connector. Press straight down until it clicks.

> **Never power the board without the LoRa antenna attached.** Transmitting without it can damage the radio.

2. Check battery polarity before plugging in: the **red wire must go to the + pad** next to the 2-pin SH1.25 battery socket. Some batteries ship wired backwards, and reversed polarity can kill the board. If reversed, don't force it — run on USB for now.
3. Plug in the battery. Fit the board into the case with the screen behind the clear window and route the antenna through its slot. The battery is often too thick for the case — if it won't close, don't force it (that's how screens crack). Leave the back off or tape the battery outside.
4. Label the boards "A" and "B" with tape.

## 2. Flash MeshCore (each board)

1. Windows only: if the board doesn't appear as a COM port, install the free **Silicon Labs CP210x** USB driver.
2. Open **flasher.meshcore.io** in Chrome or Edge.
3. Choose **Heltec V3** (may be listed as Heltec WiFi LoRa 32 V3) → **Companion (Bluetooth)** → newest version.
4. Plug in board A. If the flasher can't connect, enter boot mode:
   - Unplug, hold **PRG/BOOT**, plug back in, release after 1–2 seconds; or
   - While plugged in: hold PRG, tap RST, release PRG.

   A blank screen in boot mode is normal.
5. First time on a new board: click **Erase**, then **Flash**, and pick the serial port when the browser asks. Stay plugged in until "Flashing complete."

> When **updating** later, **skip Erase** — erasing wipes the node's identity.

6. If it doesn't reboot on its own, press **RST**. The screen should show a node name or Bluetooth PIN.
7. Repeat for board B.

## 3. Pair each board with a phone

1. In the MeshCore app, add a device over Bluetooth, pick the board, and enter the **6-digit PIN shown on its screen**. The PIN changes on every restart unless you set your own.
2. Optional: Gear icon → Bluetooth settings → Custom → enter a fixed PIN → reboot and reconnect.

## 4. Radio settings (both boards must match)

1. Name each node. Everyone on the mesh can see it, so avoid your full name.
2. Tap **Choose Presets** → **USA/Canada**, or match your local Portland mesh's settings.

| USA/Canada preset | Value |
|---|---|
| Frequency | 910.525 MHz |
| Bandwidth | 62.5 kHz |
| Spreading factor | SF7 |
| Coding rate | CR5 |

3. Location is optional — skip it to stay off public maps.
4. Save with the check mark, then **Reboot** and reconnect.

## 5. Test

1. Pair board A with one phone and board B with another (or reconnect one phone to each in turn).
2. On each board, send an **Advert** so they discover each other.
3. Open board B from contacts and send a direct message. Start in the same room, then walk apart to check range.

## Tips & troubleshooting

- **More range:** reflash one board as a **Repeater** and mount it high on steady power at the shop or house. Repeaters are managed from a companion in the app, not paired directly.
- **Battery:** the 1100 mAh cells last only a few hours — plan on USB power in the truck.
- **Boot loop / blank screen after flashing:** flash again. If it still fails, return it (Amazon holiday returns run to Jan 31, 2027).
- **Browser can't see the board:** try another data cable/USB port, install the CP210x driver, use boot mode.

Sources: nodakmesh.org/setup/heltec-v3 · docs.louisianamesh.org/setting-up-meshcore-companion · kevsrobots.com/learn/meshcore · wiki.meshamerica.com. Prepared Nov 14, 2026.